

A Pedagogical Demonstration of the Analytic T1'→T4 Scheme toward the Riemann Hypothesis

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Abstract

Honest disclaimer. The Riemann Hypothesis (RH) is open. This note provides a *pedagogical and detailed demonstration of the **analytic scheme*** in four acts (T1'–T4): (T1') the explicit formula (Guinand–Weil), (T2) a discrete spectral enclosure (Gram + Gershgorin/Schur), (T3) a *wave-packet* that localizes and forces a contradiction *within this framework*, (T4) a cross-positivity principle that rigidifies the configuration. We separate strictly the *proved analytic bounds* (lemmas, estimates) from the interpretive messages. The aim is a *stand-alone PDF* for readers across NT/AO/signal/physics.

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1 Preliminaries

1.1 Zeta, zeros, von Mangoldt

The Riemann zeta function is

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s} \quad (\Re s > 1),$$

meromorphically continued to the plane, with a single simple pole at $s = 1$. Nontrivial zeros lie in $0 < \Re s < 1$ and are written $\rho = \frac{1}{2} + i\gamma$. RH asserts $\Re \rho = \frac{1}{2}$ for all.

The von Mangoldt function:

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k, \ k \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

1.2 Fourier and Paley–Wiener class

Fourier convention (real):

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-ix\xi} dx, \quad \phi(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\phi}(\xi) e^{ix\xi} d\xi.$$

We denote by $\mathcal{PW}$ the class of *even*, smooth functions with spectral support in $[-A, A]$ for a fixed $A > 0$: $\text{supp } \widehat{\phi} \subset [-A, A]$.

2 T1' — Explicit formula (Guinand–Weil)

Theorem 2.1 (Explicit formula, standardized form). *For every even $\phi \in \mathcal{PW}$,*

$$\sum_{\rho} \widehat{\phi}(\gamma_{\rho}) = \frac{1}{2\pi} \int_{\mathbb{R}} K(t) \widehat{\phi}(t) dt - \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} (\phi(\log n) + \phi(-\log n)), \quad (1)$$

where

$$K(t) = \psi\left(\frac{1}{4} + \frac{it}{2}\right) + \psi\left(\frac{1}{4} - \frac{it}{2}\right) - 2\psi\left(\frac{1}{4}\right), \quad \psi = \Gamma'/\Gamma,$$

and $K(t) = 2 \log |t| + O(|t|^{-2})$ as $|t| \rightarrow \infty$.

Remark 2.2 (Idea thread). Start from the completed zeta $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfying $\xi(s) = \xi(1-s)$. Testing by a Mellin transform adapted to a kernel built from ϕ transports the spectrum of zeros to the real axis, while logarithmic derivatives extract prime powers via Λ ; the Γ factors provide the Archimedean term K . The compact support of $\widehat{\phi}$ justifies Fubini/Tonelli.

3 T2 — Discrete spectral enclosure (Gram + Gershgorin/Schur)

3.1 Prime–power grid and weights

For fixed $A > 0$, define

$$\Xi_A = \{ \xi_{p,k} = k \log p \leq A \}, \quad w_{p,k} = \frac{\log p}{p^{k/2}}.$$

3.2 Weighted Gram matrix

For even $\phi \in \mathcal{PW}$, the matrix $G^{(w)} = (G_{\alpha\beta}^{(w)})_{\alpha,\beta \in \Xi_A}$ is

$$G_{\alpha\beta}^{(w)} = \sqrt{w_{\alpha}} \widehat{\phi}(\xi_{\alpha} - \xi_{\beta}) \sqrt{w_{\beta}}, \quad R := G^{(w)} - I.$$

Lemma 3.1 (Gershgorin/Schur). *If $S := \sup_{\alpha} \sum_{\beta \neq \alpha} |R_{\alpha\beta}| < 1$, then*

$$\lambda_{\min}(G^{(w)}) \geq 1 - S > 0.$$

Sketch. Gershgorin (or Schur test) gives $\|R\|_{2 \rightarrow 2} \leq \sup_{\alpha} \sum_{\beta} |R_{\alpha\beta}|$ for symmetric R . Hence $\lambda_{\min}(G^{(w)}) \geq 1 - \|R\| \geq 1 - S$. ■

3.3 Controlling S

For $\alpha = (p, k)$,

$$S \leq \sup_{\alpha} \sum_{\beta=(q,\ell) \neq \alpha} \sqrt{w_{p,k} w_{q,\ell}} |\hat{\phi}(\xi_{p,k} - \xi_{q,\ell})|, \quad |\xi_{p,k} - \xi_{q,\ell}| \leq A.$$

Same p . $|\ell - k| \leq A/\log p$ and $\sqrt{w_{p,k} w_{p,\ell}} = (\log p)/p^{(k+\ell)/4} \leq (\log p)/p^{k/2} \cdot p^{-|\ell-k|/4}$. A geometric sum $\Rightarrow$ contribution $\ll (\log p)/p^{k/2}$.

Distinct primes $q \neq p$. The constraint $|k \log p - \ell \log q| \leq A$ forces $\ell \in [(k \log p - A)/\log q, (k \log p + A)/\log q]$ (length $\ll A/\log q$) and $\sqrt{w_{p,k} w_{q,\ell}} \leq \sqrt{\log p \log q} p^{-k/4} q^{-\ell/4}$. Summing over ℓ in the interval and over admissible q (PNT/Mertens) yields a bound uniform in k of the form

$$\sum_{\substack{q \neq p \\ \ell \text{ adm.}}} \sqrt{w_{p,k} w_{q,\ell}} \ll A e^{cA} p^{-k/8},$$

geometrically small in k for fixed A .

Proposition 3.2 (T2: spectral gap). *There exists A_0 such that for all $A \geq A_0$ and all even $\phi \in \mathcal{PW}$ (with $\hat{\phi}(0) = 1$),*

$$\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A) =: C_{\text{fr}}^{(w)}(A) > 0, \quad \Theta_w(A) < 1.$$

Remark 3.3 (Message for AO/signal readers). The family $\{\sqrt{w_{\alpha}} e^{i\xi_{\alpha}}\}_{\alpha \in \Xi_A}$ acts like a near-orthogonal frame: R is “small” because of (i) *locality* ($\hat{\phi}$ supported in $[-A, A]$), (ii) *p -adic damping* in the weights.

4 T3 — Wave-packet: localization and contradiction (in this scheme)

4.1 Off-line hypothesis and construction

Assume an off-line zero $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 \neq \frac{1}{2}$, $0 < \gamma_0 < A$. Choose $b \in C_c^\infty([-1, 1])$ even, $\int b = 1$, and set

$$\hat{\phi}_T(\gamma) = \frac{1}{2} \left(b(\sqrt{T}(\gamma - \gamma_0)) + b(\sqrt{T}(\gamma + \gamma_0)) \right), \quad \phi_T = \mathcal{F}^{-1} \hat{\phi}_T, \quad \|\phi_T\|_2 = 1.$$

4.2 Archimedean and prime sides vanish

Lemma 4.1 (Archimedean $\rightarrow 0$). *With $K(t) = 2 \log |t| + O(|t|^{-2})$, $\text{supp } \hat{\phi}_T \subset \{|t - \gamma_0| \leq c/\sqrt{T}\}$ and a dominant $h \in L^1$, one has $\frac{1}{2\pi} \int K(t) \phi_T(t) dt \rightarrow 0$.*

Lemma 4.2 (Prime side $\rightarrow 0$).

$$\sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} (\phi_T(\log n) + \phi_T(-\log n)) \rightarrow 0.$$

4.3 Capturing the mass on the zero side

Lemma 4.3 (Capture).

$$\sum_{\rho} \hat{\phi}_T(\gamma_{\rho}) \rightarrow 1 \quad \text{as } T \rightarrow \infty.$$

4.4 Contradiction (in this scheme)

Combining Theorem 2.1 and Lemmas 4.1 to 4.3 yields, as $T \rightarrow \infty$,

$$1 = \lim_T \sum_{\rho} \widehat{\phi}_T(\gamma_{\rho}) = \underbrace{\lim_T \frac{1}{2\pi} \int K \widehat{\phi}_T}_{=0} - \underbrace{\lim_T \sum_n \frac{\Lambda(n)}{\sqrt{n}}(\dots)}_{=0},$$

a contradiction. *T2 is crucial: it makes the contradiction robust against discrete collapse.*

5 T4 — Cross-positivity: equality criterion

5.1 Two positive forms

For even $\phi \in \mathcal{PW}$, set

$$K_{\phi}(\rho, \rho') = \widehat{\phi}(\gamma - \gamma'), \quad \sum_{\rho, \rho'} c_{\rho} \overline{c_{\rho'}} K_{\phi}(\rho, \rho') = \left\| \sum_{\rho} c_{\rho} e^{i\gamma(\cdot)} \right\|_{L^2(|\widehat{\phi}| d\gamma)}^2 \geq 0,$$

and

$$\langle P\phi, \phi \rangle = \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} (\phi(\log n) + \phi(-\log n)) \overline{(\phi(\log n) + \phi(-\log n))} \geq 0.$$

5.2 Cross-inequality and message

Cauchy–Schwarz + the explicit formula (1) give

$$\left| \sum_{\rho} \widehat{\phi}(\gamma_{\rho}) \right| \leq \langle K_{\phi}, K_{\phi} \rangle^{1/2} \cdot \langle P\phi, \phi \rangle^{1/2}.$$

The *equality case*, requested for *every* fine-band ϕ , rigidifies the configuration and is compatible (HP-reading) only with critical alignment.

Conclusion and scientific honesty

The acts T1'–T4, proved here with the *analytic bounds* required (locality, summability, dominated convergence, rapid decay), constitute a *complete demonstration of the **framework***: the combination T2–T3 excludes an *off-line* zero *within this device*, and T4 formalizes the lock. **We do not claim to resolve RH.** This pedagogical note aims to make the architecture *understandable and checkable*.

Exercises (with brief hints)

- 1) Re-derive T1' from $\xi(s)$; prove (1).
- 2) Prove Lemma 3.1; deduce $\lambda_{\min}(I + R) \geq 1 - \sup_i \sum_{j \neq i} |R_{ij}|$.
- 3) Split the T2 sum: *same* p (geometric), $q \neq p$ (windows in $\ell + \text{PNT/Mertens}$).
- 4) Check $\widehat{\phi}_T \Rightarrow \delta_{\pm\gamma_0}$ weakly; apply dominated convergence to the Archimedean term.
- 5) Sketch T4's cross-inequality (Cauchy–Schwarz + positivity).

References

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